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OPERATIONAL MANUAL
FOR
TEGMINE - TELESCOPIC CONVEYOR
510/714/6123

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1 SAFETY – Tegmine 510/714/6123

1.1 General information

The instructions for use should be read thoroughly before installation and start of the machine. Special attention should be paid to the information regarding safety and correct installation. During normal use it is important that service and inspection intervals are observed.

Only trained staff should operate the system.
Non-authorized persons should not have access to the system.

1.2 Limited use

The information and warnings in this document are only valid if the machine remains in the same condition as manufactured and delivered by Armstrong and on the condition that the user carries out the necessary maintenance and changes defective parts immediately with original spare parts.
The machine must only be used for the products for which it was designed and approved to convey.
The machine must not be exposed to climatic conditions beyond the limits specified in these directions for use or to any other overload.

1.3 Qualified staff

The service and operation staff must have sufficient qualifications and training for this work as indicated below:

Training or education that enables the staff to read and understand wiring diagrams, estimate calculations and installation data for electric connection, and to make the installation according to the valid directives.
Training or education, which provides the staff with knowledge of standards regarding usual safety on use and service of this machine.

Training or education, which enables the staff to work with programmable control units and adjustable frequency inverters or similar control system in case such systems form part of the machine.
First aid training.

NOTE! Non-observance of the safety requirements and lack of maintenance can in the worst cases cause personal injury and damage the machinery and products.

1.4 Marking

Danger
If the necessary safety precautions are not taken, death, severe personal injury or damage to the equipment may occur.
Warning
If the necessary safety precautions are not taken, death, severe personal injury or damage to the equipment may occur.
Caution
If the necessary safety precautions are not taken, personal injury or damage to the equipment may occur.
Note
Important information referring to the product or parts of the directions for use to which special attention should be paid.

1.5 Maintenance and start-up

Before putting the machine into operation, the machine must be compared with the rated voltage and operation conditions at the installation site. Furthermore the machine must be checked for transport damage.

Compare power supply requirement, shown on the machine plate, with available on-site power supply.

- Supply voltage o

Supply frequency

- Rating of distribution fuse/circuit breaker in supply switchboard

Prior to connecting the power supply, all cables, electrical components, sensors, and switches must be checked for damage. Replace components and cables, if necessary.

Check for presence of water or moisture, which could damage the electrical components. Replace components, if necessary. Presence of moisture inside the electrical components seriously reduces their durability.

Before start of the machine the existing safety regulations should be followed.

WARNING: Before operating any motors, correct direction of rotation must be checked (see the arrow on the motor). Follow the directions for use regarding control and testing.

The operator must observe the signs on the machine, which give safety warnings or instructions. In case of damage, the user shall immediately replace the signs and make sure that they are not covered in any way.

1.6 Safety of operators

Danger of leaks

Operators working at and around the machines should be informed about the danger that may arise and about how to handle dangerous situations if there is a leak of the materials mentioned below.

Hydraulic oil

The media mentioned above should only be used at the pressure values and temperature ranges prescribed for the machine. If there is a risk of exceeding the values, the supply lines should be equipped with safety equipment preventing excess of the values.

Danger at movements

Staff should not be behind any safety guards on the machine or fencing when the system is operated manually or is started by means of a program.

It is prohibited to make actions with external equipment connected with the machine, e.g. connection of hoses etc., unless the machine is properly switched off and secured against unauthorised start and operation.

Safety equipment

It is the user and his/her organisation's responsibility to ensure that normal necessary safety equipment is being used. When dismantling the underside guarding plates always use gloves.

1.7 External Emergency stop

If the boom is connected/interfaced to equipment from another manufacturer, the responsibility for safe installation rests on the contractor connecting the machinery to other overall systems. This means that additional emergency stop devices should be considered and established, if necessary. It is the responsibility of the main contractor to establish emergency stop pushbuttons, wherever considered necessary. Operators must have emergency stop pushbutton within immediate reach of their normal working position. Emergency stop pushbuttons must also be considered at all other locations, where hazards may occur.

1.8 Disposal

The machine must be disposed in accordance with the correct national environmental laws and regulations.

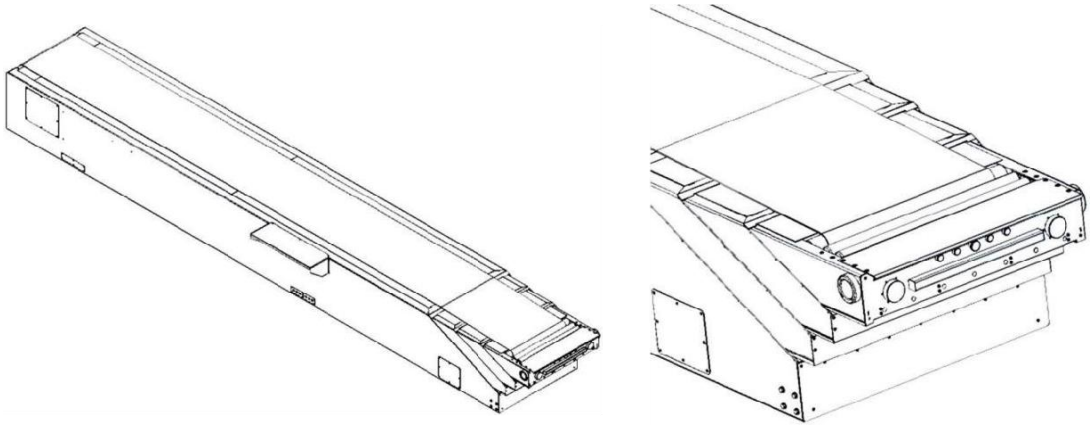
1.9 Loose fitting clothing and hair

Loose fitting clothing, jewelry; long, unbound hair can get caught in the rotating parts on machines. By keeping these things secure or removing them you can greatly reduce the risk for injury.

SYSTEM DESCRIPTION – Tegmine 510/714/6123

1.10 Introduction

Armstrong telescopic belt conveyor Tegmine 510/714/6123 is made for transport of items and parcels.
The main application of the telescopic belt conveyor is loading and unloading of trucks and containers. It is usually operated from the front of the extended sections. The workplace can be adapted to the conditions by telescopic movement without intermediate support in thetruck.



The telescopic belt conveyor has a stationary base section, section 1, which houses the telescopic boom and consists of multiple nested sections that extend out simultaneously and proportionally away from the base section. All directions are referenced as if standing at the base section and with the boom telescoping away from you.

Each boom section is supported by heavy-duty steel wheels that ride on a flat, smooth surfaced steel rail, which is welded to the bent steel side panels of the boom. Lateral control of each boom is provided in the front and rear to ensure proper alignment of the telescopic movement. Each section slider bed is a steel plate.

All sections are covered from the bottom, and there is no access to moving parts like chains, rollers, and belt, the tolerances between the sections are small in order to avoid nip points.

1.11 Options

Options for the telescopic boom conveyor:

Reversal of the belt
Motor speed controlled with frequency converter
PLC for control functions
Anti-collision strip with light
Second Joystick
Store-feed function
Photocell

Height adjustment, hydraulic

INSTALLATION AND START - Tegmine 510/714/6123

1.12 Unpacking

The telescopic boom conveyor is normally delivered wrapped in plastic.
If the telescopic boom conveyor is not to be installed within one week of delivery, the plastic must be removed or loosened in order to avoid paint damage.

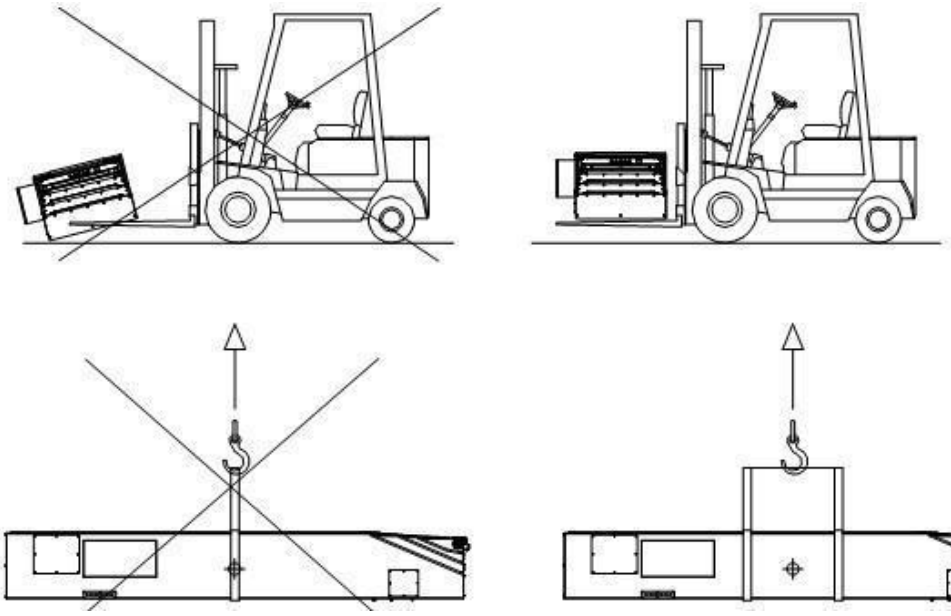
The telescopic boom conveyor should be stored indoors and protected against moisture and temperature variations.

While unpacking, use only appropriate tools to minimize the risk of damaging the conveyor. Special consideration should be given to the electrical equipment and belt.

1.13 Lifting and handling

The conveyor can be moved by using a crane, fork lift or other appropriate lifting equipment capable of lifting the load in a safe way (see illustrations below). Lift only at the base section of the conveyor.

Use spacing unit in contact areas between the conveyor and lifting device gear.
The conveyor must be in balance when lifting it, the centre of gravity is marked on all machines.
The conveyor must not be dropped. Dropping the conveyor could cause serious damage and possibly withdraw the Manufacturer's Warranty.



1.14 Mounting

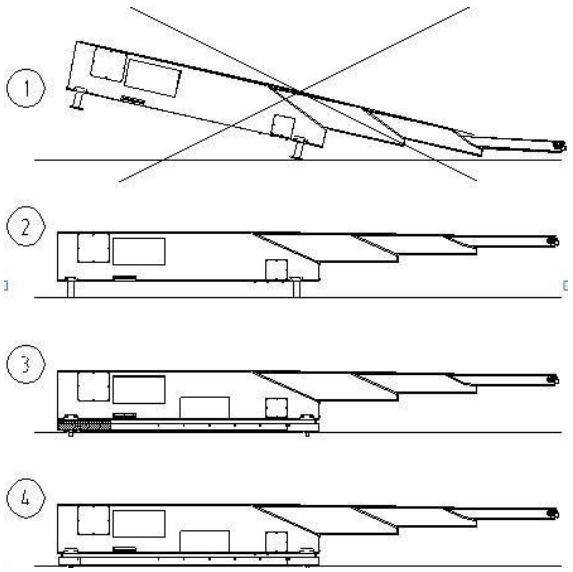
When the conveyor is placed in its final destination, and before fixing it to the foundation, you must control that the conveyor has the right level and that there is no twist from one end to the other of the base section.

A twist in the base section will cause belt-tracking problems leading to spend time on adjustment, or possibly damage the belt. If there are initial belt tracking problems check the level of the unit prior to making any belt tracking adjustments.

It is important to use appropriate levelling tools to ensure correct level installation.
The conveyor must be installed free of any torsion in the in basesection.

Anchor the unit down to the foundation by using appropriate anchors for the load forces as well as quality of foundation.
The conveyor will either be mounted directly to the foundation using the mounting feet fixed to the base section or in case where the conveyor has an optional adjustable elevation control there will be stands that should be anchored and then the conveyor will be mounted tothe stands.

If the conveyor has an option for traverse movement by an undercarriage, pos. 3 and 4 on the figure below, and that allows it to move between loading bay doors on rails embedded into the foundation, make sure that the rear hold down bar is properly attached to the conveyor and engages the rail. Some conveyors are mobile and use counter balance weights. A proper amount of weight is installed in the rear of the unit.



Warning! Before operation - make sure that the rear end of the conveyor is secured to the foundation, (2) by fixing it to the foundation, (3) by counter balance or (4) by hold down, before the booms are extended, or else (1) the conveyor will tip over and be damaged.

1.15 Connection of power supply and exchange signals

The power supply cable is to be connected to terminals on the isolator switch inside the main cabinet.
The main cabinet, which is placed on the side of the conveyor's base section, is supplied with a spare cable gland to lead the power supply cable into the cabinet.
When attaching the power supply cable, please ensure that the cable gland is tightened properly, in order to maintain a sufficient degree of protection (IP 65).
Tighten the cable gland until the rubber seal touches the cable completely, and then tighten the nut further 180°.

Supply terminals, on the main isolator, are numbered: T1, T2, and T3.
Power supply cable must be connected as follows L1 to T1, L2 to T2, L3 to T3, and PE to the earthing terminal. Please refer to section
Please ensure that the power supply is connected with CW rotary field, as the conveyor is factory tested with this. This ensures correct function and correct rotation of all electric motors.

The required power supply is 3 phases, 400 V + N + PE (Protective Earth), 50 Hz unless otherwise agreed.

> Verify that the facility supply voltage and frequency match the information provided on the machine plate, located near the main cabinet.
> Verify that the distribution fuses/short-circuit protection in the facility switchboard do not exceed the rating indicated on the machine plate. Standard is 16 Amps.

Overall system may need to be connected to the conveyor via exchange signals in order to supervise and externally control certain functions on the conveyor.
Cable for interfacing with the overall system must be connected inside the main control cabinet, to the terminal row X3. The cable must be lead through an available spare cable gland mounted in the main cabinet.
When attaching the interface cable, please make sure that the cable gland is tightened properly in order to maintain a sufficient degree of protection (IP 65).
Tighten the cable gland until the rubber seal touches the cable completely and then tighten the nut further 180°.

Some standard exchange signals are prepared, but please refer to the electrical schematics in section 10, for further details.

1.16 Start

Before taking the conveyor into use the following should be checked:

The telescopic booms are not blocked.

The belt is clear of all objects before turning on power at the main isolator switch on the main control cabinet.

Do not begin loading the conveyor before the control functions have been tested, and it has been confirmed that all functions work properly. Please refer to section 5.

The telescopic boom conveyor must be monitored during the first hours of operation. If it's not properly installed, the belt may not track correctly. If it is not correctly installed the belt could run unevenly and into the sides of the main structure of the conveyor and hereby be damaged.

During normal operation of the conveyor it is not unusual for the belt to move +/- 20 mm from side to side. If the belt moves more than 30 - 40 mm to one side, the belt should be stopped immediately and adjusted. Before adjusting the belt tracking, verify that the unit is installed in level and that the base section does not twist. The belt will not track properly if the unit is not levelled. For Belt Tracking see section

2 OPERATION – Tegmine 510/714/6123

2.1 Introduction

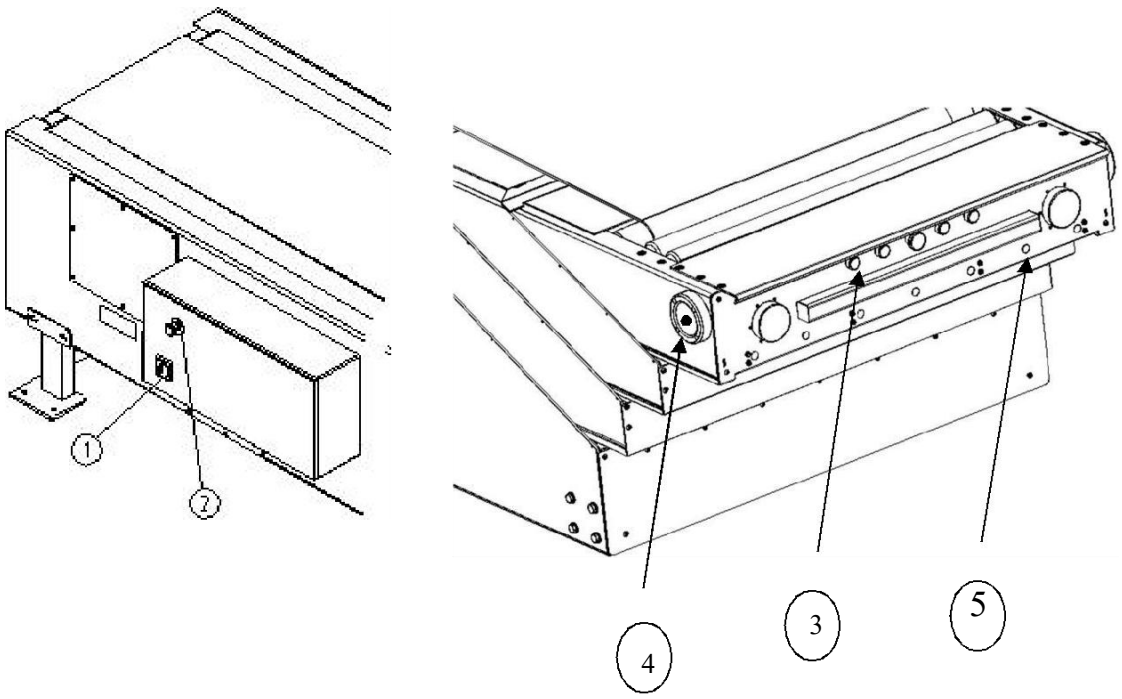
The telescopic boom conveyor has one main control cabinet placed on the side of the base section. The control on the main control cabinet typically includes emergency stop pushbutton and main isolator switch.

(1) The main isolator switch for disconnecting power to the conveyor is intended for switching off power, prior to cleaning or service. Always use a personal padlock to lock the main isolator switch in the "OFF" position, in order to prevent unintentional reestablishment of the power supply, while doing service/maintenance on the conveyor.
Please refer to section.

(2) Activating the emergency stop pushbutton inhibits all functions and cuts power for all motors. The emergency stop pushbutton maintains its position after activation and must be reset to restart the conveyor. To reset the emergency stop pushbutton, it must be turned counter clockwise. Before resetting, the operator must ensure that injured persons or damaged goods have been removed from the operating area.

The control panel for belt and boom operations is at the front end of the conveyor. The control panel typically includes emergency stop pushbutton, control buttons for belt and light, joystick, and anti-collision strip. Optional front light can be supplied and will be built in the anti-collision strip.

(3) The control buttons are used for belt control and light. By activating the "Belt Forward" pushbutton, the belt will start running in the loading direction. The "Belt Reverse" pushbutton, which is optional, will start the belt in the unloading direction. To stop the belt, the operator must activate "Belt Stop". However, it must be noted, that the belt motion can depend on exchange signals from the overall system. "Belt Forward" and "Belt Reverse" are illuminated pushbuttons, which indicate, if they have been activated, and to warn the operator, that the belt may be started by the overall system. The pushbutton for the front light is an ordinary spring-return pushbutton, which, via a latching relay, in the main cabinet, toggles the front light on and off.

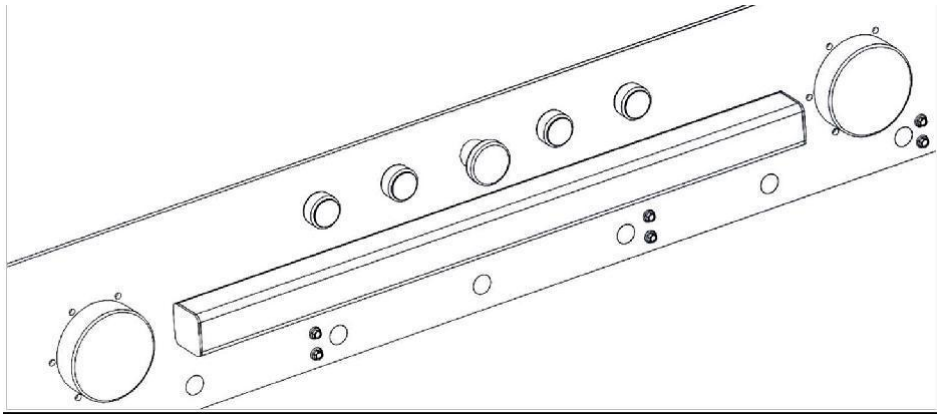


(1) Main isolator switch
(2) Emergency stop button
(3) Control buttons
(4) Joystick
(5) Anti-collision device

(4) The Joystick is used for boom length adjustment and levelling (levelling is optional). Toggling the joystick forth or back causes the boom to extend or retract. Fully retracted and fully extended is supervised by built in limit switches. As an option the conveyor can be supplied with height adjustment. Toggling the joystick up or down causes the boom to raise or lower. Extreme limits in this direction depend on the hydraulic cylinders. There are no switches to detect this. Therefore it is advisable to release the joystick as soon as upper or lower limits are reached and to avoid overloading the hydraulic pump.

(5) The anti-collision device is photoelectric induction type that mounted on the face of the boom to prevent the boom from extending into any objects, gate-doors, or personnel. When the boom extends and the anti-collision device is activated, the boom stops extending, and reverses (retracts) until the joystick is released, or the boom is fully retracted.

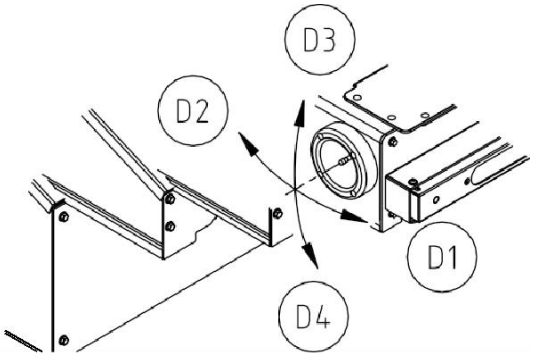
2.2 Control functions



Push button	Function
Belt Forward	Start belt
Belt Reverse (optional)	Start belt for reversing
Belt Stop	Stop belt
Light (optional)	Light on/off

If the machine is equipped with extra options please refer tosection

Joystick movement	Function
D1	Boom extend
D2	Boom retract
D3 (optional)	Boom raises
D4 (optional)	Boom lowers
Joystick at opposite side is optional	



2.3 Use

The telescopic conveyor is designed only for transport of parcels and goods, and is therefore not to be used for passenger transport.

Because of the physical structure, it is possible to make the telescopic booms swing from side to side. Breakdown or damage caused by getting the telescopic booms into a forced harmonic vibration is not Armstrong responsibility.

Do not overload the boom conveyor; it has a max. load carrying capacity of 50 kg/m. Place parcels gently on the conveyor especially in case of heavy parcels in unloading situations. The shock impact will reduce the lifetime of the conveyor's components.

Protective safety guards should only be removed for service and maintenance, and should remain in place at all other times. Damaged guards and covers are to be replaced immediately. Use local site safety power lockout procedures when removing guards.

During service and maintenance the main isolator switch must always be switched off and secured by a personal padlock. Never carry out service, while the power is switched on!! Please refer to section

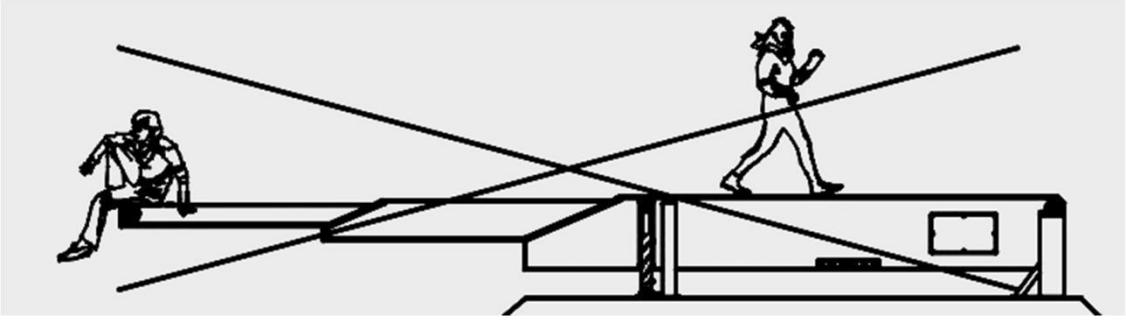
A briefing concerning correct use of the safety functions must take place, including anti-collision strip and emergency stops.

2.4 Operation Precautions

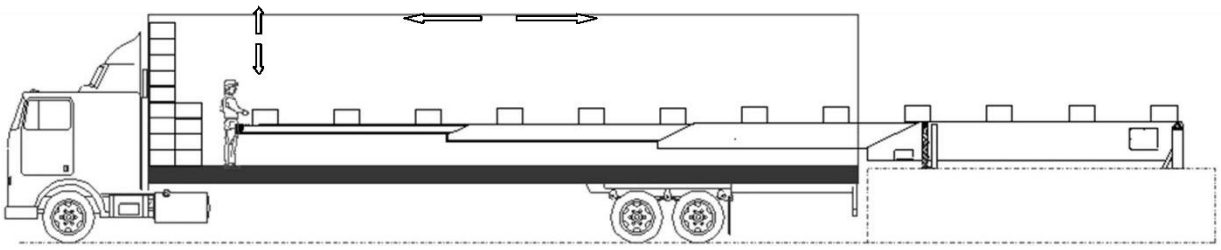
WARNING! Do not walk or crawl underneath the telescopic conveyor while in operation.

Do not place fingers under the belt while the conveyor is in operation.
Do not touch anything underneath the boom sections. Keep fingers and clothes free of anyopening. See also SAFETY

DO NOT WALK, STEP, CRAWL ON OR UNDER THE CONVEYOR



2.5 Examples of operation



The telescopic boom conveyor is in use when loading or unloading a truck. The operator controls the conveyor from the front, so it is always in the right position.
The telescopic boom conveyor is in use when loading or unloading a truck with a container. The conveyor is installed with height adjustment (optional), and hereby the operator can adapt the height of the conveyor to the required level regardless the sort of truck being loaded/unloaded. To get the best belt tracking on the conveyor the parcels must be placed on the centre of the belt. If the conveyor is equipped with a hood it is extremely important that the parcels are placed on the centre of the belt in order to avoid tracking problems.

3 Maintenance

3.1 Maintenance plan

(At 8 hours daily operation 5 days a week)

Check interval		Area	Action
Days or Hours			
Daily		Entire machine	Abnormal noises or vibrations. Other visible damage. Verify that all protective shields and guards are in place. Verify proper operation of the anti-collision strip in the front of the boom. Leakage in the hydraulic system (if equipped with hydraulics).
		Belt drive	Belttracking. Belt damage.
Monthly	160	Control system	Check all push button controls for operational responses. Test Emergency Stops. Check cables visually, flex track, drive chains for bending, cuts, or breakage.
		Rollers	Check for build-up of debris on rollers and pulleys. Occasionally string, tape, paper, etc. will get wrapped around rollers. When this is the case, stop the unit and carefully remove the debris. Telescopic support wheels and guide tracks are to be cleaned.
		Sections	Cleaning of bottom cover in the operating area in the front of the extended sections. Make sure power is disconnected in order to avoid accidents.
Every 6 months	1000	All chains	Rub off chains with a cloth dipped in oil and lubricate with grease. Verify that the chain is tight.*
			Check chains and sprockets for wear.
		Bearings	Visual check of wear on bearings.
Yearly	2000	Control system	Check if limit switches are fixed properly and adjusted.
			Check if photocells are fixed properly and adjusted (Optional).
	500	Hydraulic system	Change hydraulic oil. (Optional)
Every 2 years	10000	Gear motors	Change gear oil.
			Clean and adjust brake on telescopic gear motor, it requires special skill and knowledge.

* Verify the first time after 3 month's operation.

3.2 Cleaning

Outer surfaces and belt should be cleaned as normally with vacuum hover or a soft cloth.

Operator panel should be cleaned cautiously to avoid damaging the panel.

All sections are covered from the bottom, which means that dust and pieces from parcels like paper or tape can be caught inside the conveyor, especially in the front section for loading and unloading. It is recommended to clean this area once a month.

Clean inside areas when carrying out maintenance.

4 REMEDY OF FAULTS - Tegmine 510/714/6123

4.1 Telescopic Drive

PROBLEM	POSSIBLE REASON	CORRECTIVE ACTION
The drive motor does not move.	Check power to the motor.	
The chains do not move.	Defect sprocket.	Find the reason and use spare parts.
	Chains do not grip the sprocket.	Tighten the chain. (See telescopic movement)
Telescopic boom will not extend/retract.	Is the belt running?	Start belt motion.
	Emergency stops.	Release all emergency stops.
Telescopic boom stops errati-cally or inconsistently.	Dust accumulation on the brake.	Check stationary disc guides for dust. Remove rubber seal collar and clean with air hose.
Telescopic motion pulls to the left or right.	Loose chain.	Tighten chain.
	Chain off sprocket.	Put chain on and tighten.
	Chain broken.	Replace chain or repair link in chain.

4.2 Belt Drive

PROBLEM	POSSIBLE REASON	CORRECTIVE ACTION
The drive motor does not move.	Check power to the motor.	
The belt does not track properly.	Torsion in the conveyor.	Level the conveyor.
	Dirt on the pulleys or on the drive pulley.	Clean pulleys.
	None of the above.	Adjust the belt as described in the instruction manual.
The belt slips on the drive pulley.	The belt is blocked.	Remove the object blocking the belt.
	The belt is overloaded.	Remove the overload.
	None of the above.	Tighten the belt as described in the section 5.5.1 Belt adjustment.
The belt is damaged on the edges.	The belt is rubbing on the chain drive.	Adjust the belt as described in section 5.5.1, and if any cuts, repair the belt or replace.
Deep cuts in the belt.	Damaged by load.	Repair the belt.
	Damaged by something else.	Find and removethe damaging object, repair or replace the belt.
The belt comes apart.	Old cuts, which have not been repaired.	Remove the damaging object, repair or replace the belt.
Noise from the belt drive.	Objects are stuck on the belt path.	Remove the objects and if necessary, repair the belt.
	Noise from one of the pulleys. The bearings are damaged.	Replace the pulley or the bearings.

4.3 Electrical

PROBLEM	POSSIBLE REASON	CORRECTIVE ACTION
No motor reaction at start.	Power failure.	Check power supply.
	Wrong direction of motor rotation.	Change phase sequence in supply line.
	Motor circuit breaker switches off.	Reconnect motor overload and check cause of fault.
	Defective motor circuit breaker, contactor, or wrong wire connections.	Change defective components and check cause. Refer to section 10, for further details.
	Damaged control cable.	Check control cables for damage. Use spare wires or replace, if necessary.
	Damaged motor cable.	Check motor cables for damage, and replace, if necessary.
Motor overload trips (immediately upon activation)	Damaged motor cable Short-circuit.	Check motor cables for damage, and replace, if necessary.
	Defective motor winding.	Change motor.
	Mechanical motor blockage.	Check pump, transmission.
	Motor circuit breaker set too low.	Change setting of motor overload from nominal current to max. 10% above nominal current. Nominal current can be read on identification plate, either on motor, or inner side of the main control cabinet door.
Periodical trip of motor overload.	Motor circuit breaker set too low.	Change setting on motor circuit breaker from nominal current to max. 10% above nominal current. Nominal current can be read on identification plate, either on motor, or on the inner side of main control cabinet door.
	Periodical power failure.	Check power supply.
	Mains voltage periodically too low.	Check mains voltage.
No functions operate.	No supply voltage.	Verify main isolator switch is in ON position.
		Verify that distribution fuses/short-circuit protection in the facility switchboard are OK/switched on.

		Verify supply voltage to be as required, when measured on main isolator switch.
	No control voltage.	Check that the automatic circuit breakers are switched on.
		Check control voltage on control voltage supply unit; -G1. Replace, if necessary.
		Verify that all emergency stop pushbuttons are reset.
	Emergency stop interface.	Verify all external emergency stops are reset, or jumpers are properly mounted in interface terminals (X3:)
	Cable damage.	Check control cables for damage. Use spare wires or replace, if necessary.
	Bad connection.	Check all wire terminations, and tighten if necessary.
Belt will not run.	Motor circuit breaker –Q1 tripped.	Switch on. If the same problem occurs immediately, check the motor cable for damage, and check the motor for damage. Replace if necessary. Check if the belt can run smoothly.
	Faulted control.	Check relevant pushbuttons for damage, and replace, if necessary.
	Damaged belt motor.	Check belt motor. Replace if necessary.
Boom will not extend or retract.	Belt not running.	Verify that the belt is running. On some units telescopic movement is inhibited, while the belt is stopped.
	Motor circuit breaker –Q2 tripped.	Switch on. If the same problem occurs immediately, check the motor cable for damage, and check the motor for damage. Replace if necessary.
	Soft starter –U2 set too low.	Increase the initial torque setting on soft start.
	Soft starter –U2 does not work.	Replace soft starter.

	Faulted controls.	Check relevant pushbuttons/joystick for damage, and replace, if necessary.
	Damaged boom motor.	Check motor, replace if necessary.
Boom only retracts, even when “Extend” is activated.	Anti-collision strip stuck.	Check anti-collision strip. Due to the built in springs, it must return to neutral position, after each activation. Grease or replace if necessary.
Boom will not extend.	Damaged limit switch.	Check the boom fully extended limit switch.
	Damaged control cable.	Check control cables for damage. Use spare wires or replace, if necessary.
	Motor circuit breaker –Q2 tripped.	Switch on. If the same problem occurs immediately, check the motor cable for damage, and check the motor for damage. Replace if necessary.
Boom will not retract.	Damaged limit switch.	Check the boom fully retracted limit switch.
	Damaged control cable.	Check control cables for damage. Use spare wires or replace, if necessary.
	Motor circuit breaker –Q2 tripped.	Switch on. If the same problem occurs immediately, check the motor cable for damage, and check the motor for damage. Replace if necessary.
Boom will not raise. (For units with the hydraulic elevation option)	Motor circuit breaker –Q3 tripped.	Switch on. If the same problem occurs immediately, check the motor cable for damage, and check the motor for damage. Replace if necessary.
	Incorrect motor rotation diretion for pump.	Check for proper motor rotation. Change phase sequence, for pump motor, if necessary.
	Damaged or incorrectly adjusted hydraulics.	See hydraulic trouble shooting section.

	Damaged hydraulic motor or pump assembly.	Check motor, and check for leakage. Replace if necessary. (See hydraulic trouble shooting section).
	Faulted controls.	Check relevant pushbuttons/joyst ick for damage, and replace, if necessary.
Boom will not lower.	Solenoid valve not operating. (Fitted on hydraulic station)	Check mechanical operation of the solenoid valve. Replace if necessary.
Undercarriage does not move. (Optional)	Motor circuit breaker –Q4 tripped.	Switch on. If the same problem occurs immediately, check the motor cable for damage, and check the motor for damage. Replace if necessary.
	Proximity switchdamaged. (Optional for positioning)	Check proximity switch for damage. Also check cable for switch. Replace if necessary.
	Faulted controls.	Check relevant pushbuttons for damage, and replace, if necessary.
	Damaged boom motor.	Check motor, replace if necessary.

Escalation Matrix
for
Armstrong Machine Builders Pvt Ltd

Help Desk	Escalation Level 1	Escalation Level 2	Escalation Level 3	
Call Log in Customer Relationship Manager (CRM)	**Nashik Office** **Mr. Abhijeet Jadhav** (Assistant Service Manager) Contact No :- 9673008242 abhijeet.jadhav@armstrongltd.com service @armstrongltd.com	**Nasik Office** **Mr. Vijay Khairnar** (Service Manager) Contact :- 9673009060 vk@armstrongltd.com service @armstrongltd.com	**Pune Office** **Mr.Dilip Deshpande** (Technical Director) Contact :- 9673009062 dd@armstrongltd.com service @armstrongltd.com	